

L1 Review Questions

Review Question:



Recall this JPEG compressed image. In the simplified JPEG process from Lecture 1, which operation directly discards image information?

- A. Dividing the image into 8x8 blocks
- B. Representing the image blocks with a discrete cosine transform (DCT)
- C. Setting some DCT weights to zero
- D. Reordering the DCT weights using a zig-zag scan

What Does Demosaicking Produce?

A Bayer sensor contains $M \times N$ photosites. Each photosite measures only one colour component. After demosaicking, the image is represented as an $M \times N \times 3$ RGB array. Which statement best explains this?

- A. The sensor directly measured all three colour components at every location.
- B. Demosaicking estimates the two missing colour components at each location using nearby measurements.
- C. Each sensor measurement is copied into the red, green, and blue channels.
- D. Demosaicking triples the spatial resolution of the image.

Answer: B: Demosaicking increases the number of represented colour values but not the number of spatial sample locations. The two missing channel values are estimated, not directly measured.

What Artifact Remains?

A grayscale image is reduced from 256 possible intensity levels to only 16. Which visual artifact is most likely to appear?

- A. Visible bands appear across smoothly varying regions.
- B. Halos appear around sharp edges.
- C. Random speckle appears in uniform regions.
- D. Regular 8x8 block boundaries become visible.

Answer: A. A smoothly varying region requires more intensity resolutions to represent and lower intensity levels leads to banding. Also called false contouring